CS-3009 Computer Architecture-Spring 2026

Assignment#3

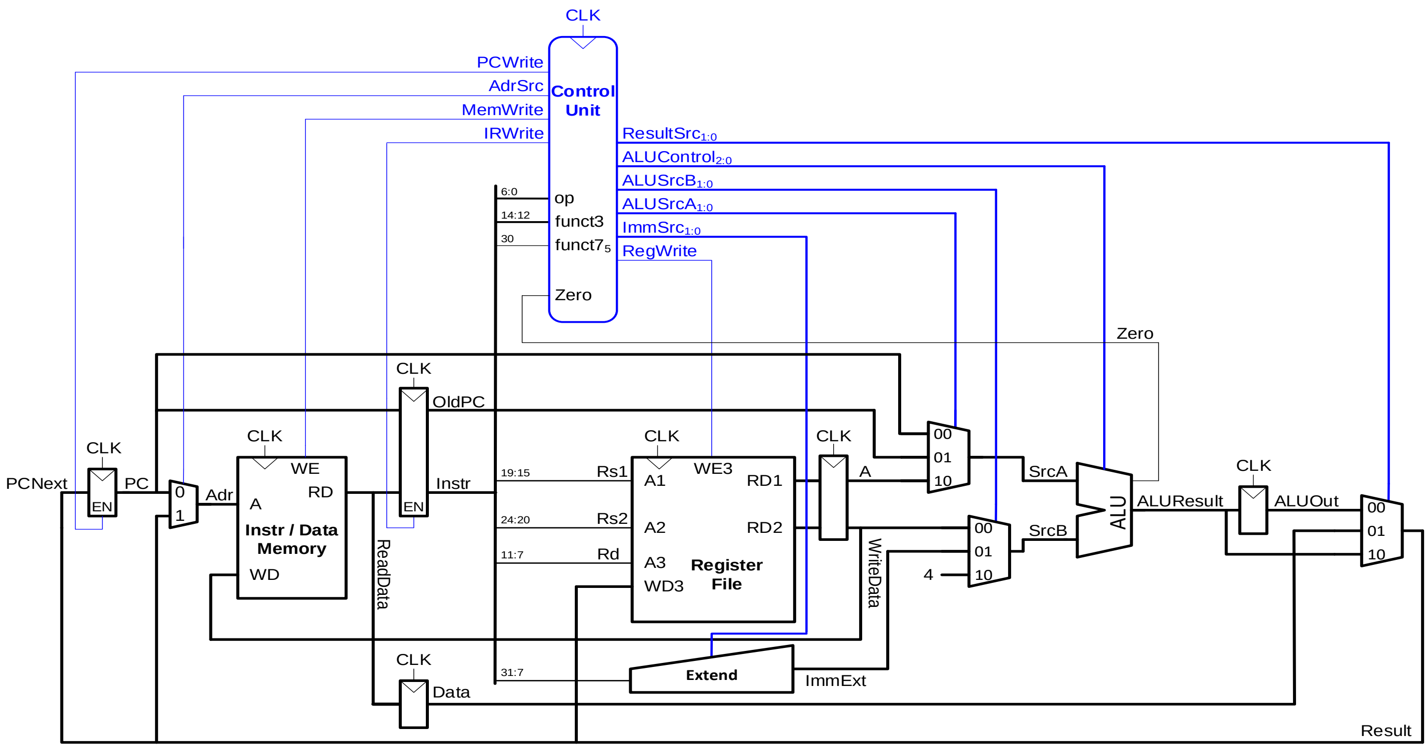
Due Date and Time: 11th May, 2026 (08:30 AM)

Instructions:

1. Late assignment will not be accepted
2. Only handwritten attempt will be graded, i.e., printed attempts will not be graded
3. There will be no credit if the given requirements are changed
4. Your solution will be evaluated in comparison with the best solution
5. Please mention the question number and its part (if any) before writing down its solution. Use the same conventions used in the assignment's document
6. Don't plagiarize. Plagiarism would lead to straight zero in the assignment regardless of the percentage copied
7. Also submit scanned copies of your assignment on GCR. 20% penalty for not submitting it on GCR
8. Your evaluation is based on the quiz from this assignment. So be ready for quiz on the submission day

1. For the instruction given below fill out control signals and highlight dataflow for the given instruction in diagram given below **(5)**

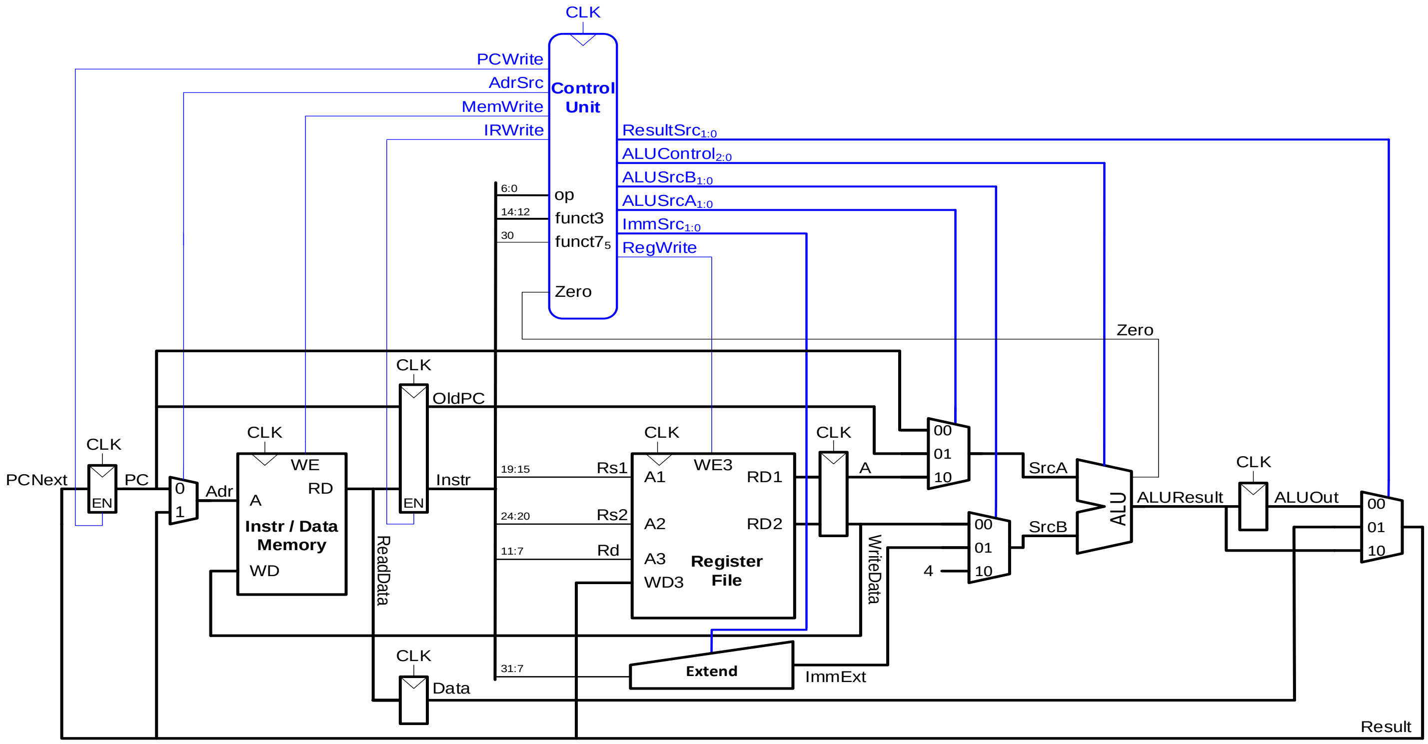
Add s1, s5, s3



Opcode	INSTRUCTION	REG WRITE	IMMSRC	ALU SCR	MEM WRITE	RESULT SRC	BRANCH	PC SCR	ALUOP
	Add s1, s5, s3								

2. For the instruction given below fill out control signals and highlight dataflow for the given instruction in diagram given below **(5)**

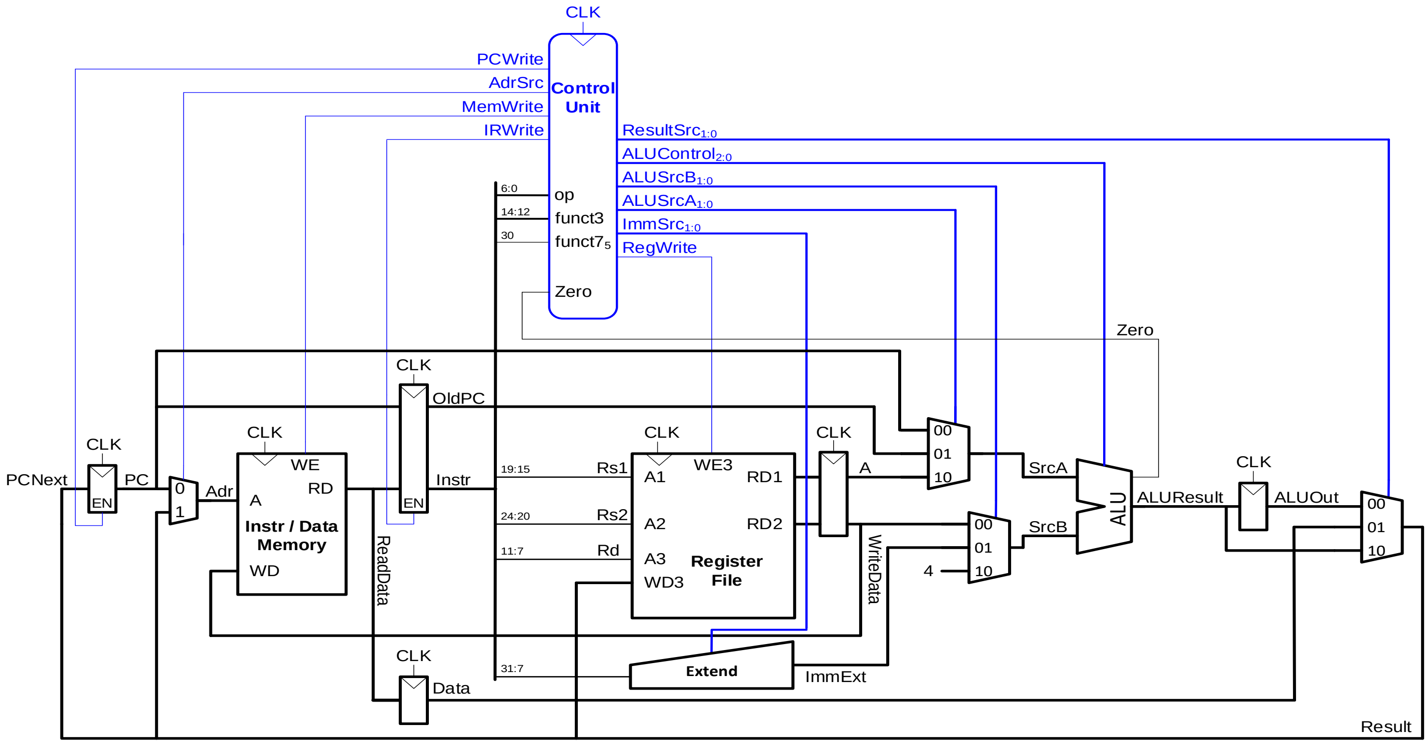
andi s1, s5,-5



Opcode	INSTRUCTION	REG WRITE	IMMSRC	ALU SCR	MEM WRITE	RESULT SRC	BRANCH	PC SCR	ALUOP
	Andi s1,s5,-5								

3. For the instruction given below fill out control signals and highlight dataflow for the given instruction in diagram given below **(5)**

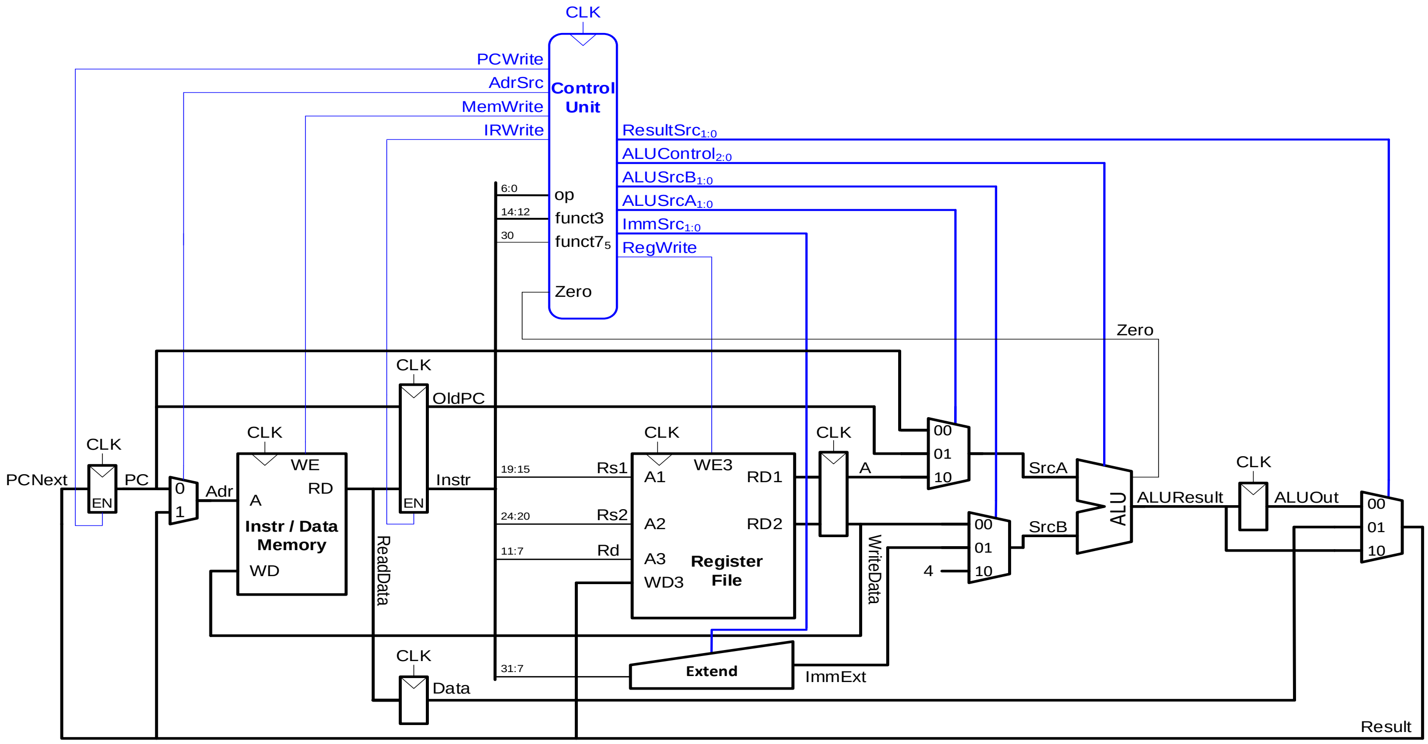
`lw t0,4(t4)`



Opcode	INSTRUCTION	REG WRITE	IMMSRC	ALU SCR	MEM WRITE	RESULT SRC	BRANCH	PC SCR	ALUOP
	lw t0,4(t4)								

4. For the instruction given below fill out control signals and highlight dataflow for the given instruction in diagram given below **(5)**

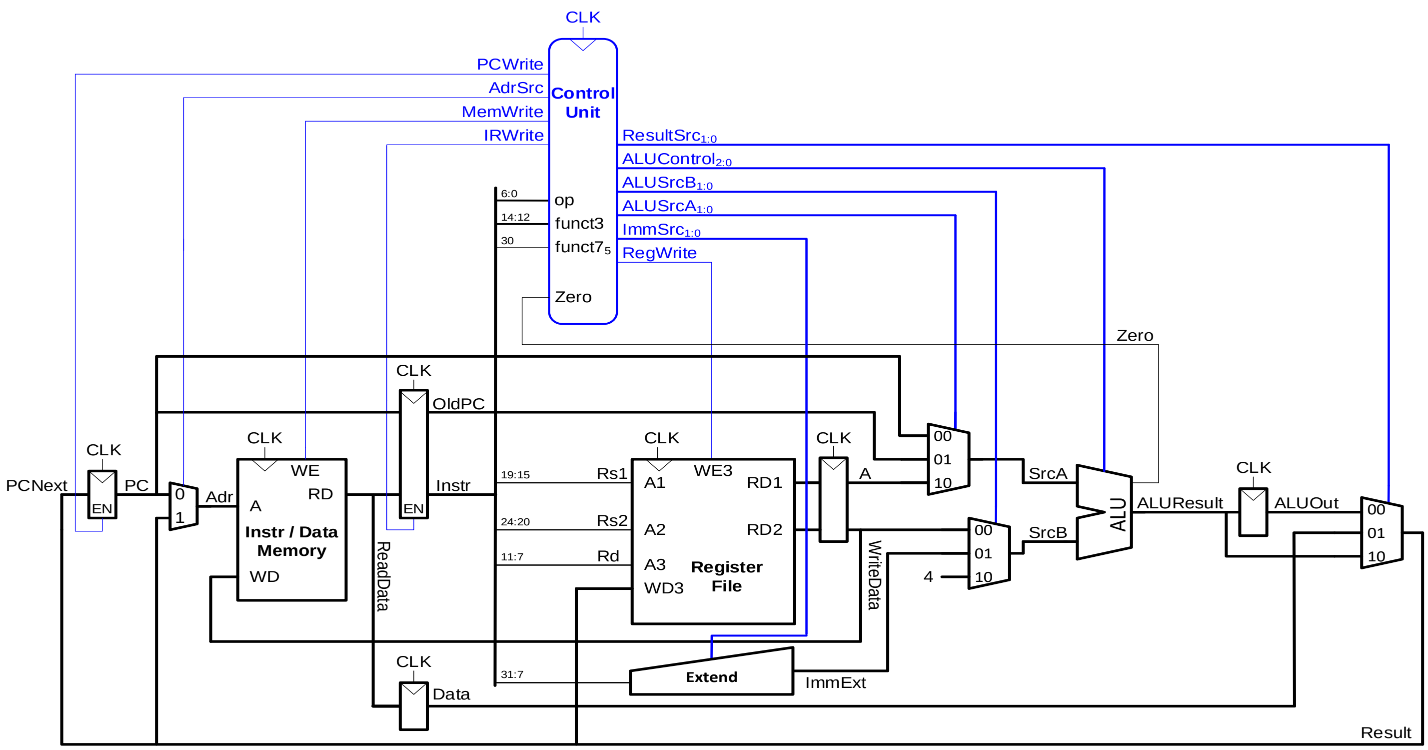
sw a0,4(s2)



Opcode	INSTRUCTION	REG WRITE	IMMSRC	ALU SCR	MEM WRITE	RESULT SRC	BRANCH	PC SCR	ALUOP
	sw a0,4(s2)								

5. For the instruction given below fill out control signals and highlight dataflow for the given instruction in diagram given below **(5)**

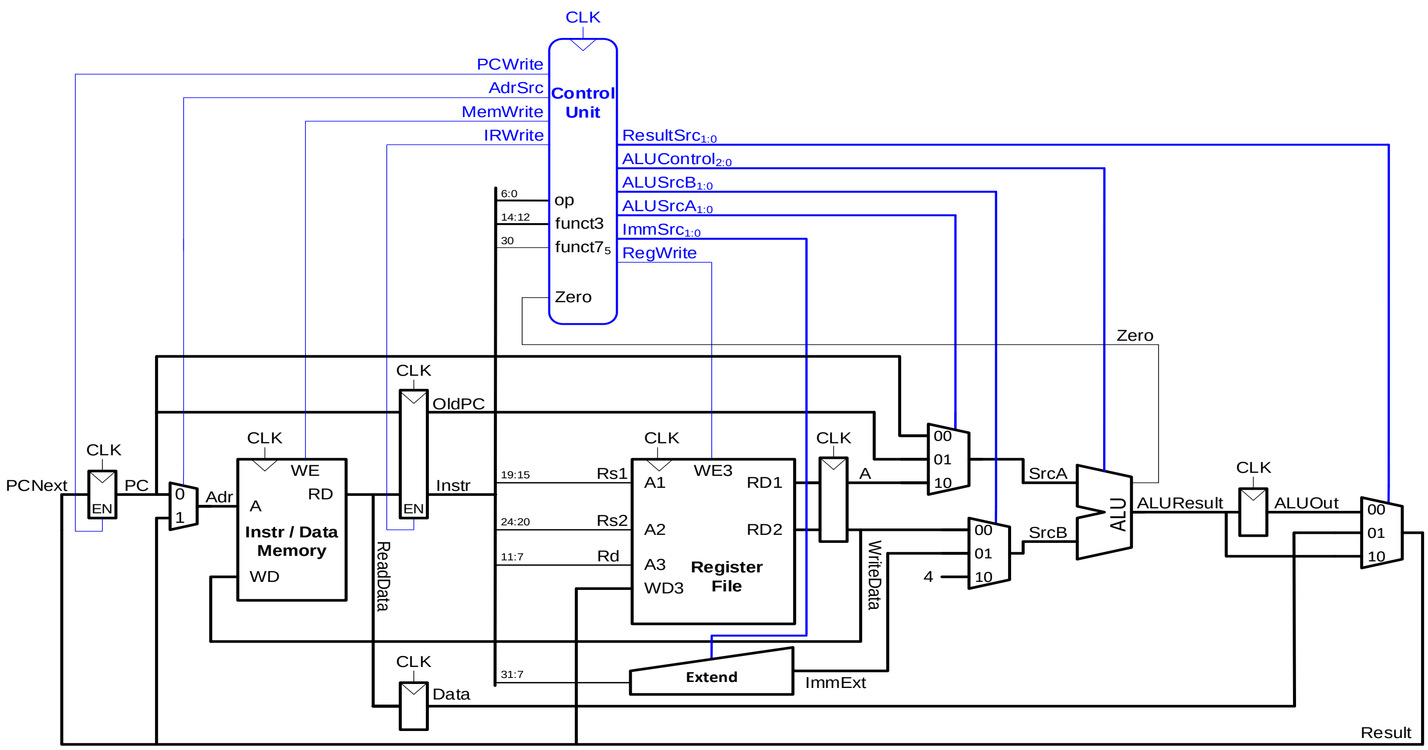
beq s5, t0, L1



Opcode	INSTRUCTION	REG WRITE	IMMSRC	ALU SCR	MEM WRITE	RESULT SRC	BRANCH	PC SCR	ALUOP
	beq s5, t0, L1								

6. For the instruction given below fill out control signals and highlight dataflow for the given instruction in diagram given below **(5)**

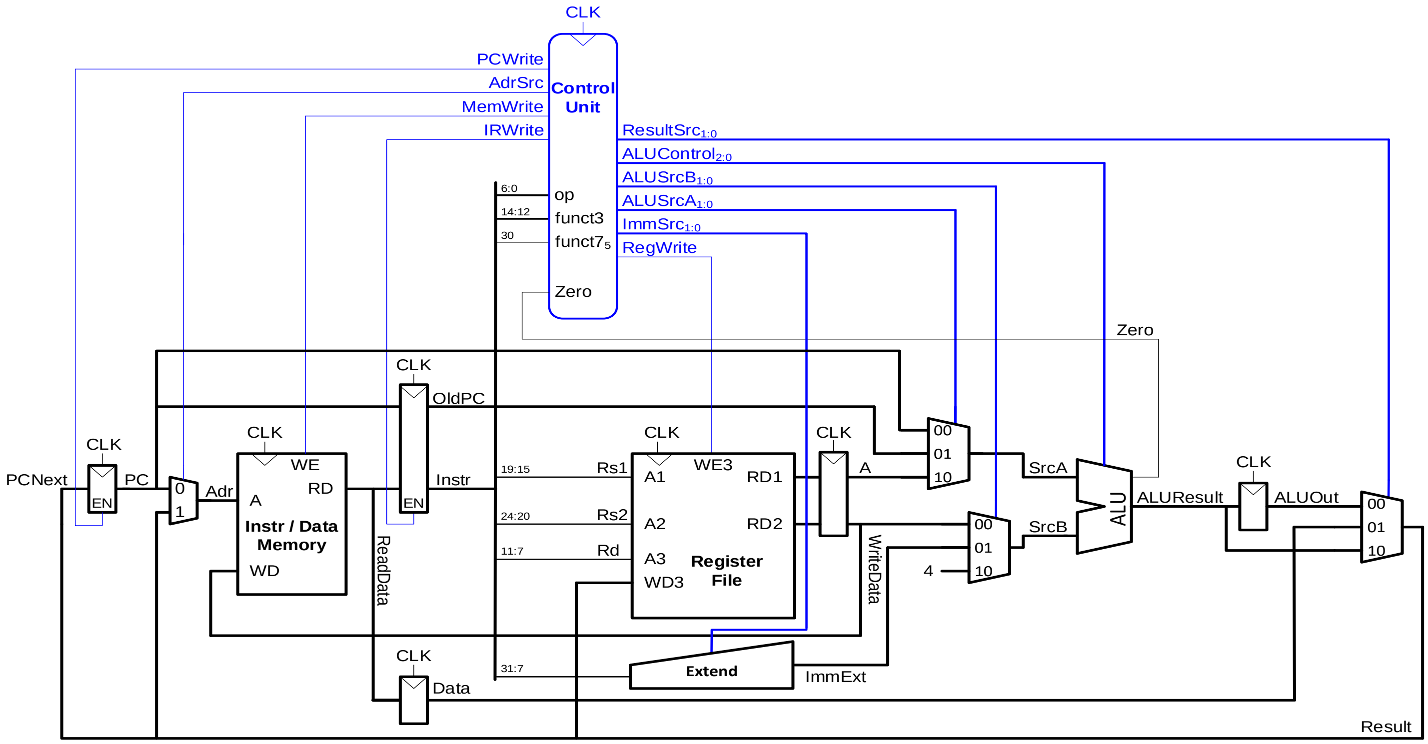
Jal ra, funct1



Opcode	INSTRUCTION	REG WRITE	IMMSRC	ALU SCR	MEM WRITE	RESULT SRC	BRANCH	PC SCR	ALUOP
	Jal ra, funct1								

7. Modify the given dataflow for the following instruction fill out control signals and highlight dataflow for the instruction **(5)**

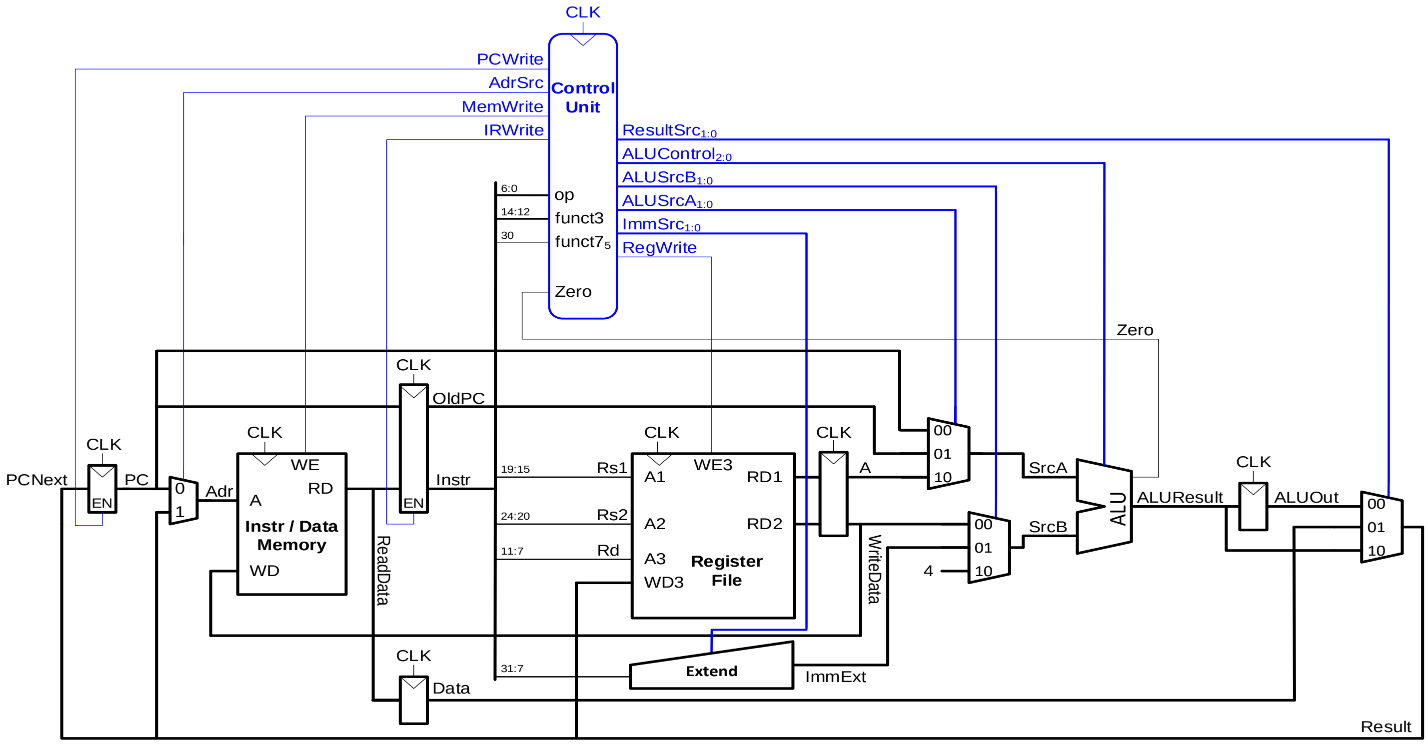
Jr ra



Opcode	INSTRUCTION	REG WRITE	IMMSRC	ALU SCR	MEM WRITE	RESULT SRC	BRANCH	PC SCR	ALUOP
	Jr ra								

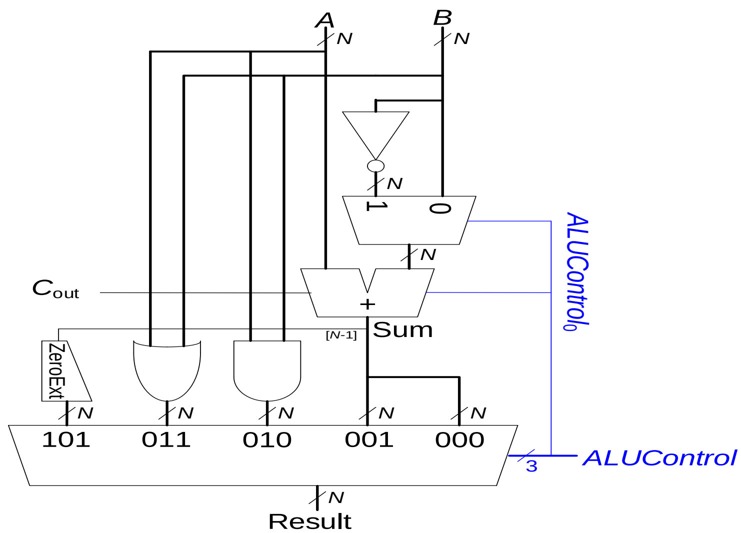
8. Modify the given dataflow for the following instruction fill out control signals and highlight dataflow for the instructions. You are supposed to add a comparator in the ALU might require more control signals. Modify ALU and ALU table in cooperate comparator in ALU **(5)**

Bgt s2, s1, **label1**



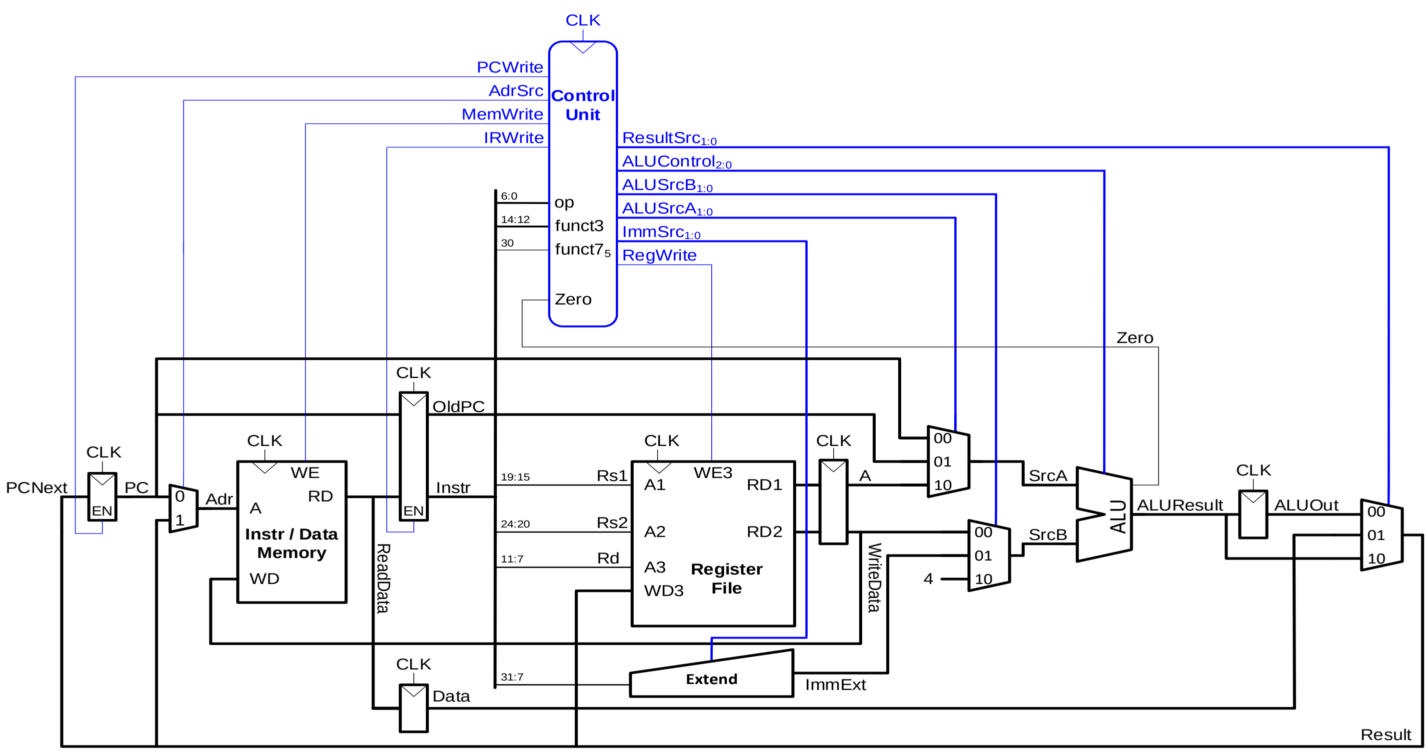
Opcode	INSTRUCTION	REG WRITE	IMMSRC	ALU SCR	MEM WRITE	RESULT SRC	BRANCH	PC SCR	ALUOP
	Bgt s2, s1, **label1**								

ALUControl$_{2:0}$	Function
000	add
001	subtract
010	and
011	or
101	SLT
110	Bgt/ble



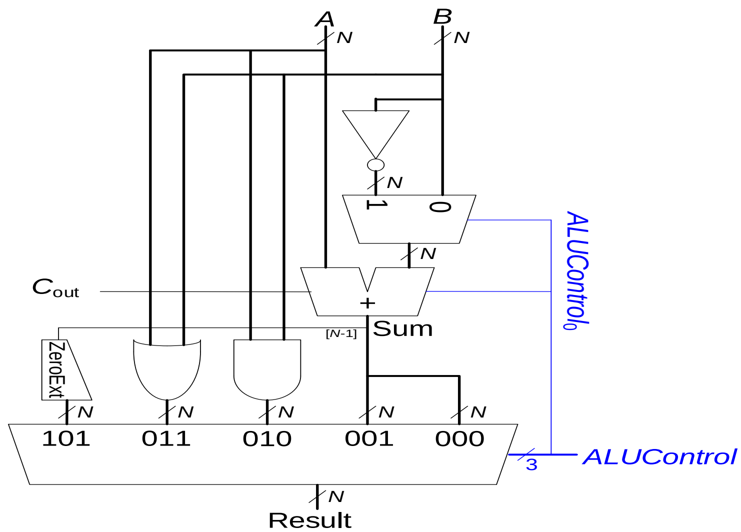
9. Modify the given dataflow for the following instruction fill out control signals and highlight dataflow for the instructions. You are supposed to add a shifter in the ALU might require more control signals. Modify ALU and ALU table incooperate shifter in ALU **(5)**

Slli s2, s7,5



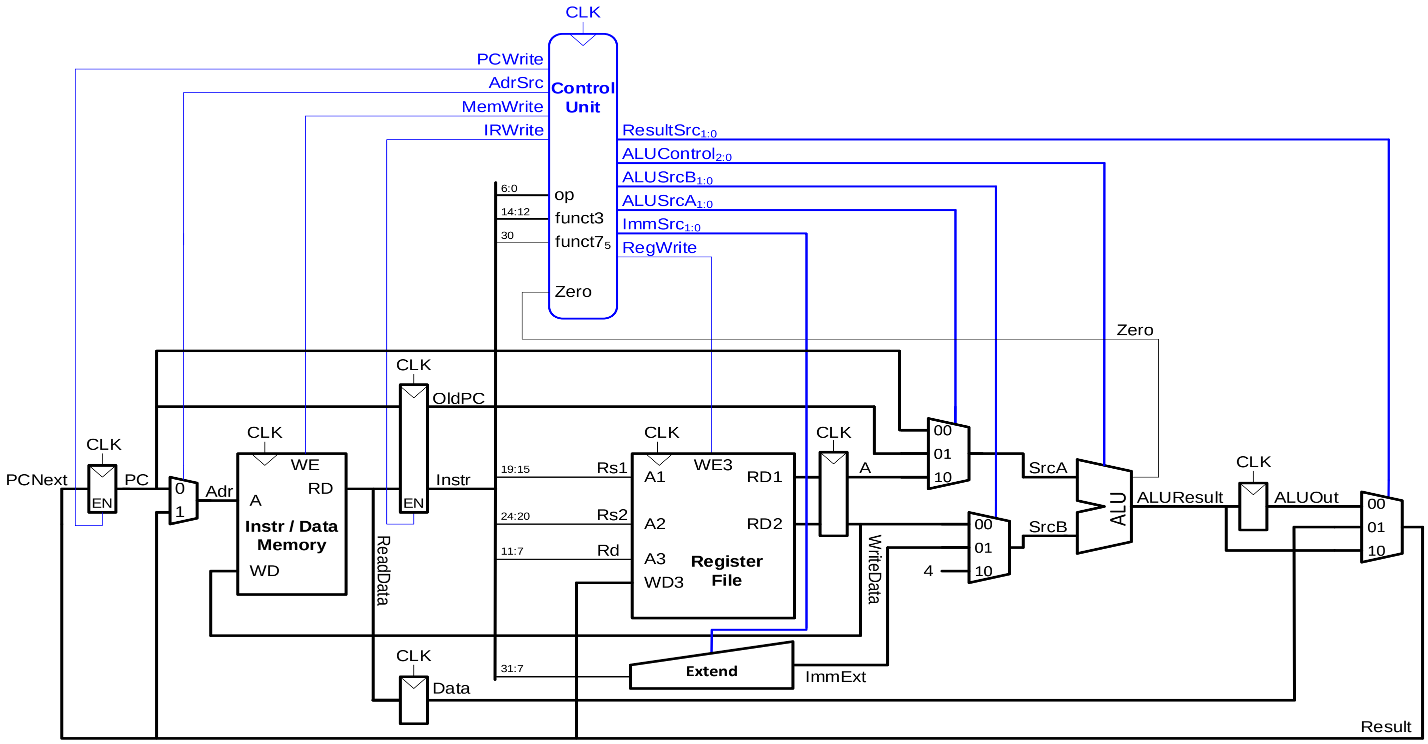
Opcode	INSTRUCTION	REG WRITE	IMMSRC	ALU SCR	MEM WRITE	RESULT SRC	BRANCH	PC SCR	ALU OP
	Slli s2, s7,5								

ALUControl$_{2:0}$	**Function**
000	add
001	subtract
010	and
011	or
101	SLT
110	Bgt/ble
111	SLLi



10. Modify the given dataflow for the following instruction fill out control signals and highlight dataflow for the instruction. Use your mind. **(5)**

Lb s4,0x1F(s4) ;load byte



Opcode	INSTRUCTION	REG WRITE	IMMSRC	ALU SCR	MEM WRITE	RESULT SRC	BRANCH	PC SCR	ALUOP
	Lb s4,0x1F(s4)								

11. Modify the given dataflow for the following instruction fill out control signals and highlight dataflow for the instruction **(5)**

```
sh s4,23(t0)      ;store half word
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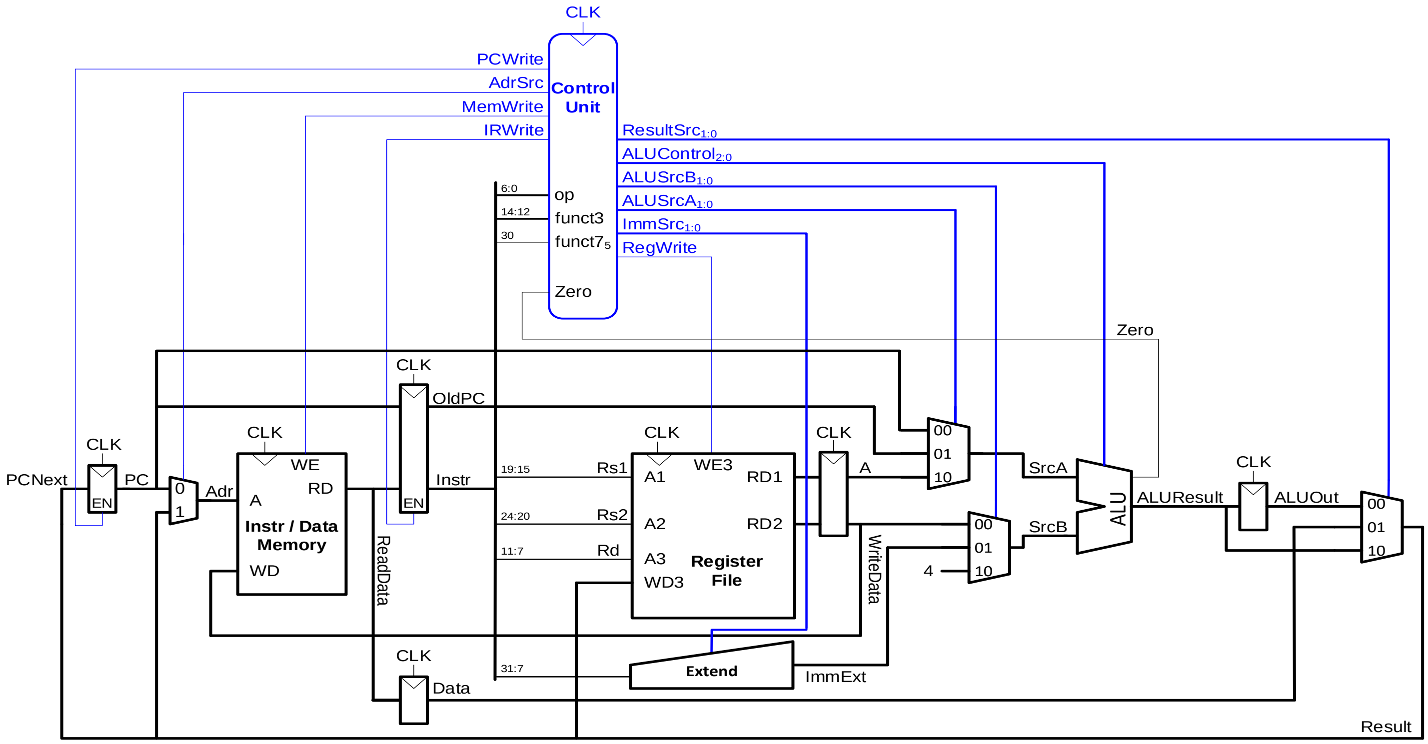
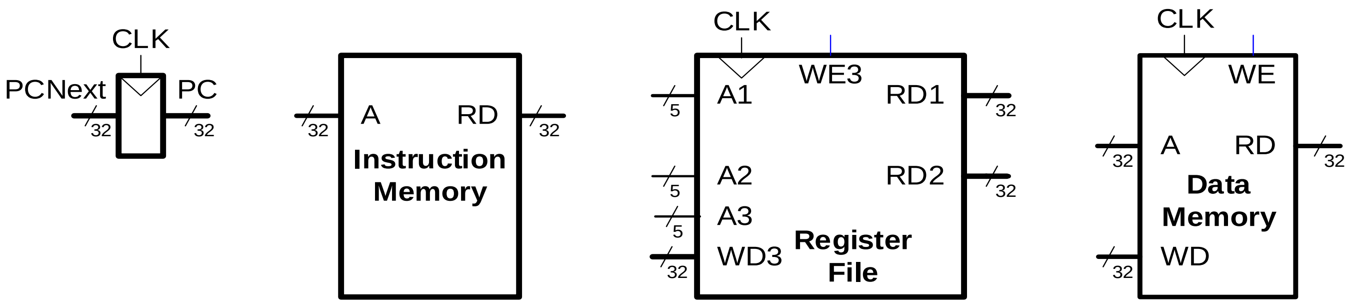


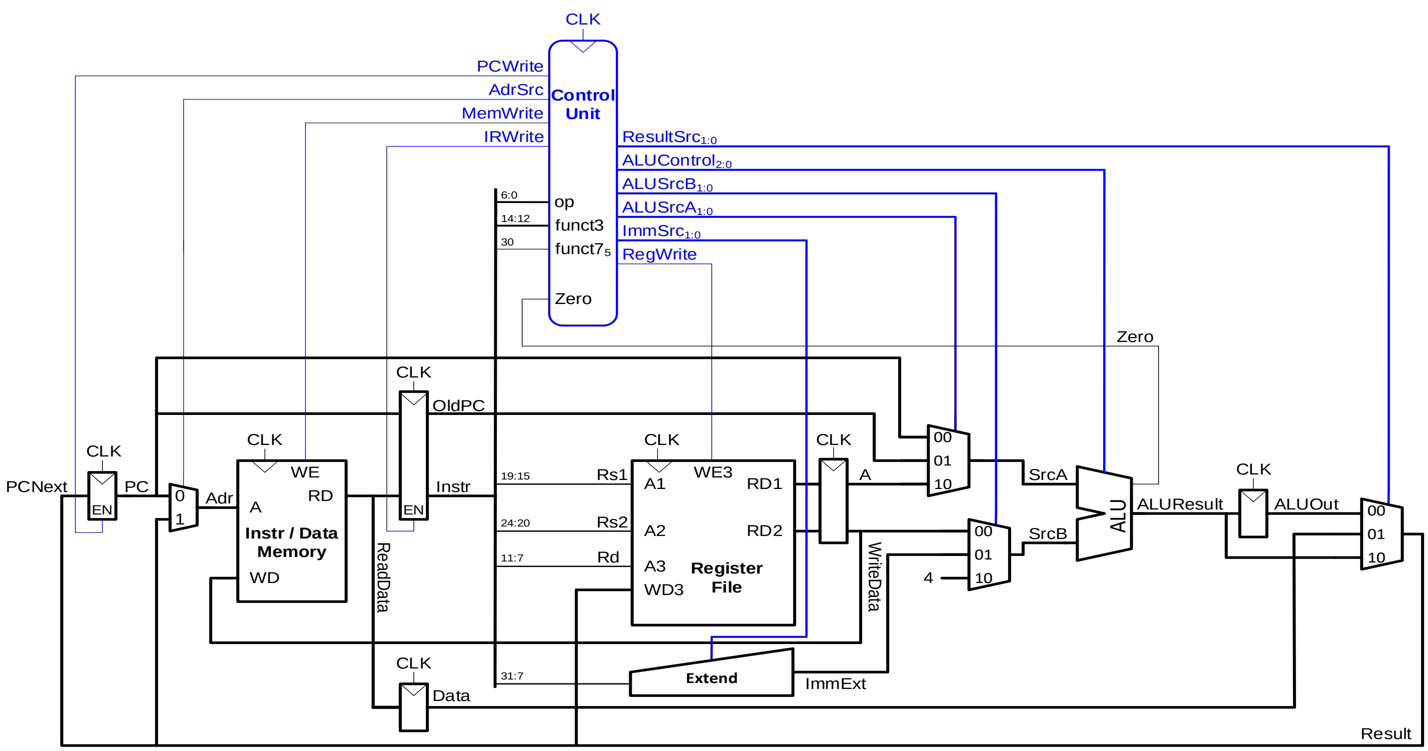
Opcode	INSTRUCTION	REG WRITE	IMMSRC	ALU SCR	MEM WRITE	RESULT SRC	BRANCH	PC SCR	ALUOP
	sh s4,23(t0)								

12. Consider the following component required for **single cycle** path complete the path for following instruction **(5)**

Lui x1, 0x56789



13. You are supposed to cooperate above given instruction Lui x1, 0x56789 into given Datapath given below **(5)**



14. Mention and Justify changes made to above dataflow for **Lui x1, 0x56789** instruction? Fill given control table for instruction **(5)**

INSTRUCTION	REG WRITE	IMMSRC	ALU SCR	MEM WRITE	RESULT SRC	BRANCH	PC SCR	ALUOP
LUI								